



Big Data Analytics

Big Data Analytics Tools

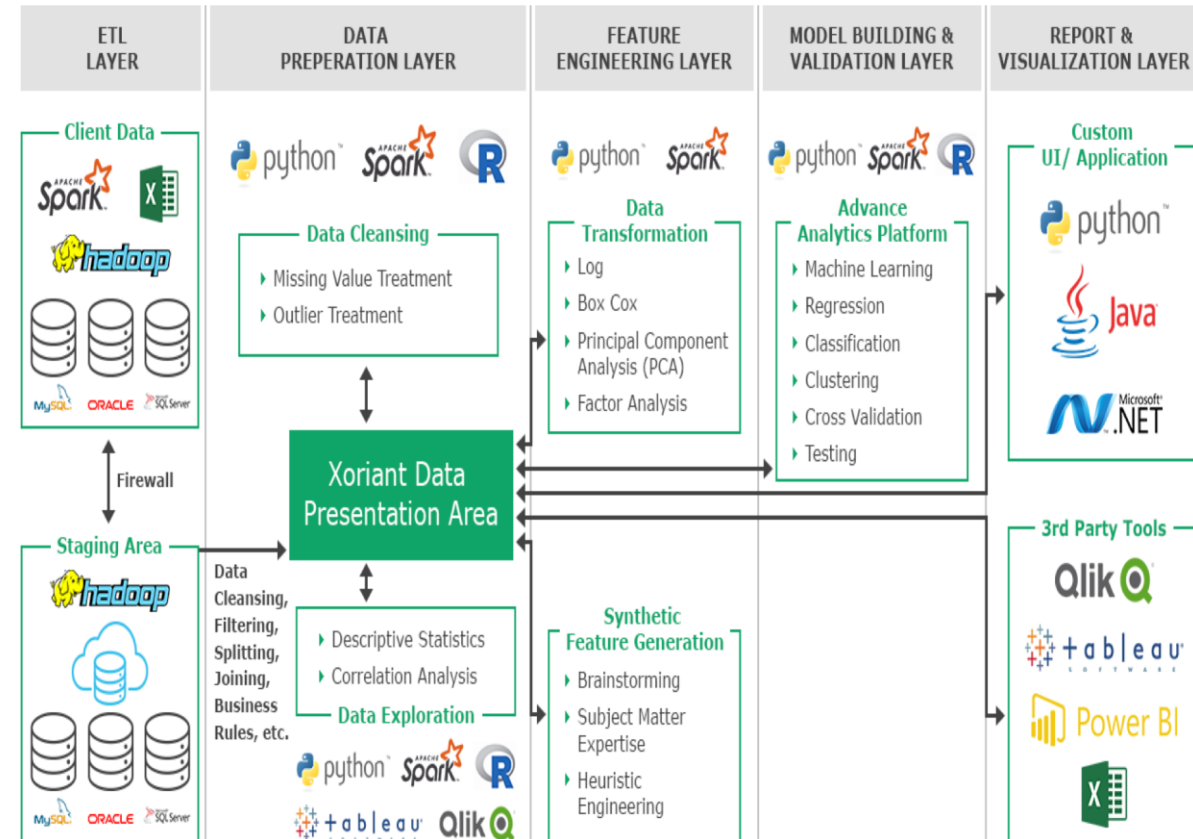
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 - ✓ Hadoop Distributed File System (HDFS)
 - ✓ MapReduce
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 - ✓ Other Components of Hadoop Ecosystem
- ❖ Apache Spark

Big Data Analytics Tools

- ❖ Excel
- ❖ Tableau
- ❖ R
- ❖ Python
- ❖ Apache Hadoop
- ❖ Apache Spark
- ❖ Apache Storm
- ❖ Power BI

- ❖ SharePoint
- ❖ Gephi
- ❖ MATLAB
- ❖ Weka
- ❖ SAS



Big Data Analytics Tools-Cont'd

- ❖ There is an increasing number of cloud-based big data analytics tool:
 - Amazon SageMaker,
 - Google BigQuery ML,
 - Google Dataproc,
 - Microsoft Azure ML,
 - IBM Cloud,
 - NVIDIA GPU Cloud,
 - Google Collaboratory

Hadoop

- Hadoop is an **open-source** software framework used for **storing** and **processing** Big Data in a **distributed** manner on large clusters of **commodity hardware**.
- Hadoop was developed based on the paper written by Google on MapReduce and Google Distributed File System (GDFS).
- In Hadoop, data resides in a distributed file system which is called as a **Hadoop Distributed File system (HDFS)**.
- The processing model is based on '**Data Locality**' concept wherein computational logic is sent to cluster nodes (server) containing data.

Hadoop-as-a-Solution

Storing huge amount of data:

- HDFS provides a distributed way to store Big Data.
- Data is stored in blocks in **DataNodes**.
- Hadoop follows **horizontal** scaling instead of vertical scaling.
- In horizontal scaling, you can add new nodes to HDFS cluster on the run as per requirement, instead of increasing the hardware stack present in each node.

Hadoop-as-a-Solution-Cont'd

Storing the variety of data:

- Stores **all kinds** of data whether it is structured, semi-structured or unstructured.
- In HDFS, there is **no pre-dumping schema** validation.
- It also follows **write once and read many** model.
- Due to this, you can just write any kind of data once and you can read it multiple times for finding insights.

Hadoop-as-a-Solution-Cont'd

Processing big data faster:

- **Move processing unit to data** instead of moving data to processing unit.
- Instead of moving data from different nodes to a single master node, the processing logic is sent to the nodes where data is stored.
- So that each node can process a part of data in parallel.
- Finally, all of the intermediary output produced by each node is merged together and the final response is sent back to the client.

Hadoop-as-a-Solution-Cont'd

Reliability:

- If one of the machines fails, another machine will take over the responsibility and work in a reliable and fault tolerant fashion.
- Hadoop infrastructure has **inbuilt fault tolerance features**.

Economical:

- Hadoop uses **commodity hardware** (like your PC, laptop).
- For example, in a small Hadoop cluster, all your DataNodes can have normal configurations like 8-16 GB RAM with 5-10 TB hard disk and Xeon processors.
- So, Hadoop ownership and maintenance is economical because Hadoop is an **open source** software.

Hadoop-as-a-Solution-Cont'd

Scalability:

- Hadoop has the inbuilt capability of integrating seamlessly with cloud-based services.
- So, if you are installing Hadoop on a cloud, you don't need to worry about the scalability factor because you can go ahead and procure more hardware and expand your setup within minutes whenever required.

Flexibility:

- Hadoop is very flexible in terms of ability to deal with all kinds of data.
- Data can be of any kind and Hadoop can store and process them all, whether it is structured, semi-structured or unstructured data.

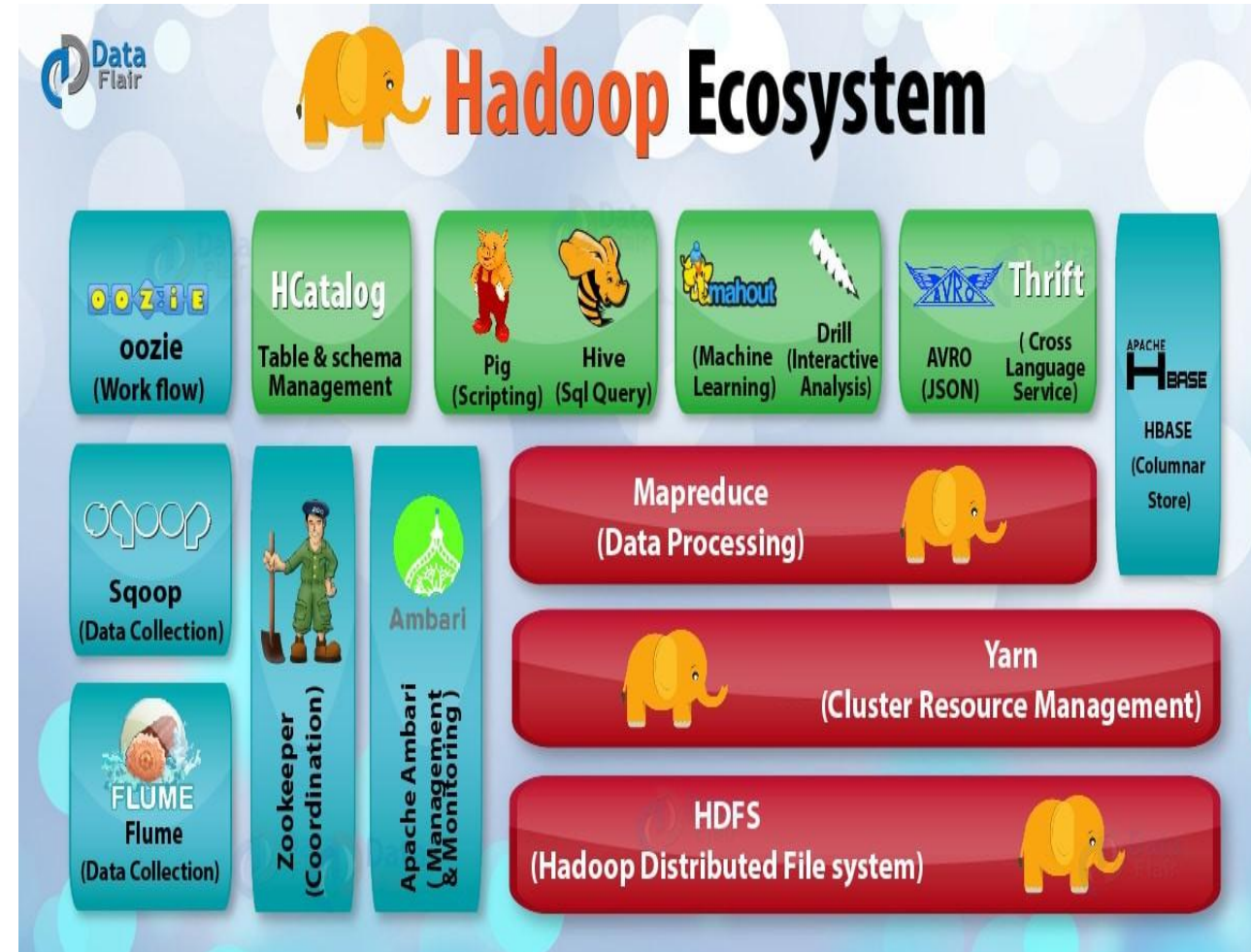
Hadoop Ecosystem and Components

❖ Hadoop is a suite of services:

- Ingesting
- Storing
- Analyzing
- Maintaining

❖ Hadoop Components:

- Hadoop Distributed File System (HDFS).
- Yet Another Resource Negotiator (YARN).
- MapReduce.



Hadoop and Its Ecosystem

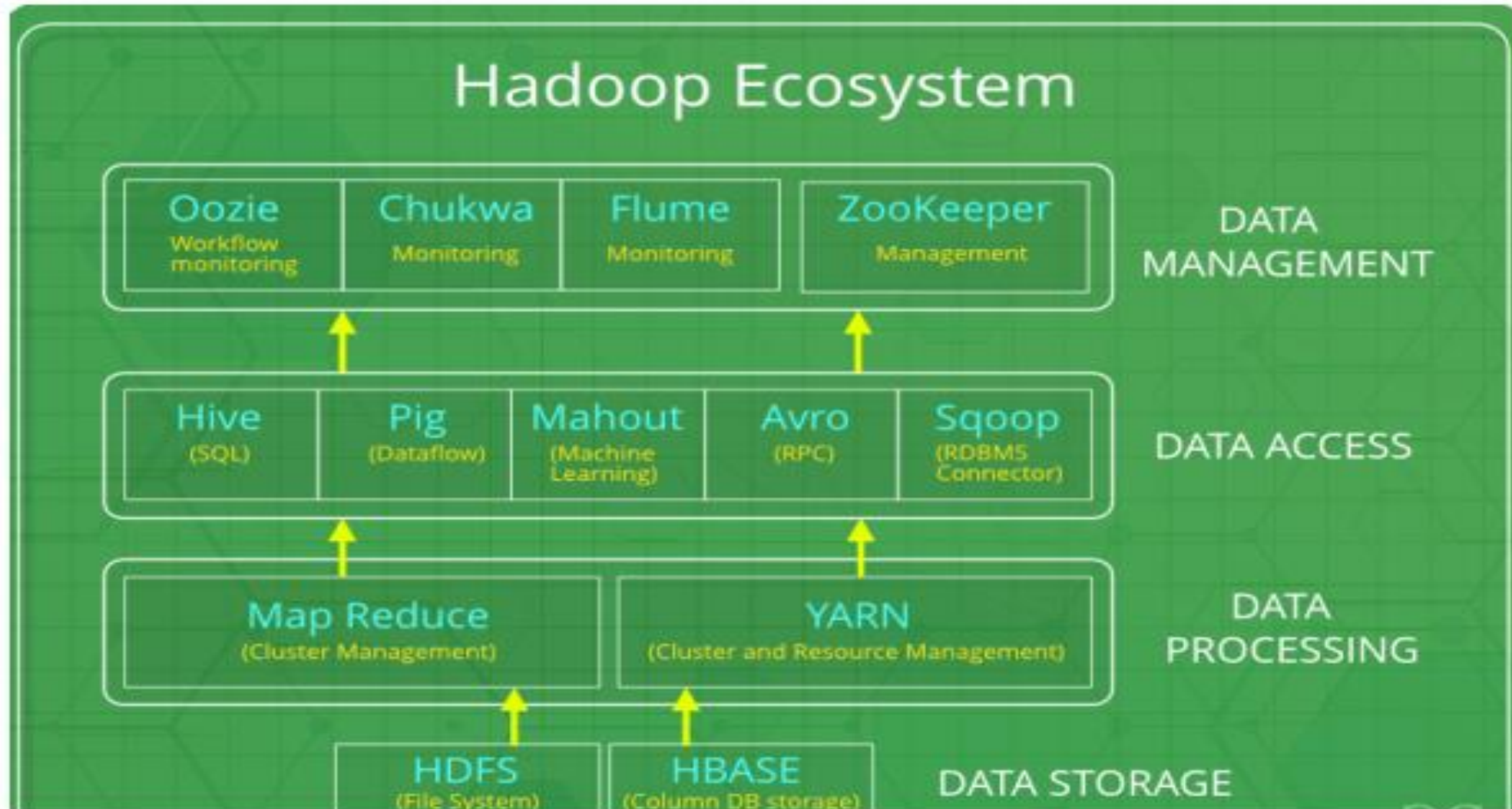
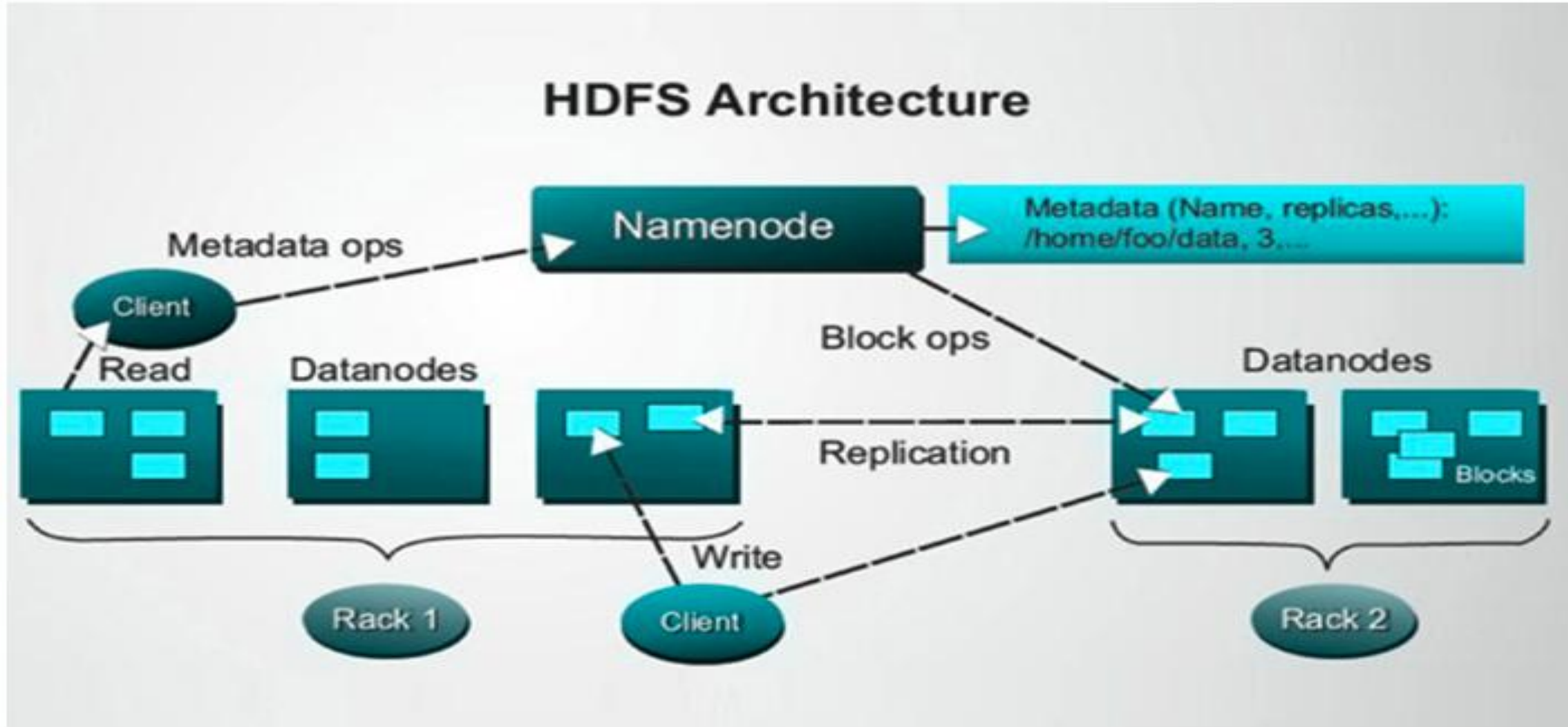


Figure 2.5 Hadoop Ecosystem

Hadoop Distributed File System (HDFS)

- HDFS is the primary storage system of Hadoop.
- HDFS is a java based file system that provides scalable, fault tolerance, reliable and cost efficient data storage for big data.
- It's inspired by the Google File System(GFS).
- Distribute large data file into blocks.
- Blocks are managed by different nodes in the cluster
- HDFS is a distributed filesystem that runs on commodity hardware.
- Hadoop interact directly with HDFS by shell-like commands.
- HDFS has two major components - NameNode and DataNode.

HDFS Architecture



Hadoop Distributed File System (HDFS)-Cont'd

NameNode:

- Also known as *Master* node.
- NameNode does **not** store actual data or dataset.
- NameNode stores **Metadata** i.e. number of blocks, their location, on which Rack, on which Datanode the data is stored and other details.
- It consists of files and directories.

Tasks of HDFS NameNode

- Manage file system namespace.
- Regulates client's access to files.
- Executes file system execution such as naming, closing, opening files and directories.

Hadoop Distributed File System (HDFS)-Cont'd

DataNode:

- Also known as **Slave**.
- Responsible for storing actual data in HDFS.
- Performs **read and write** operation as per the request of the clients.
- Replica block of Datanode consists of 2 files on the file system.
 - The first file is for data.
 - The second file is for recording the block's metadata.
- HDFS Metadata includes checksums for data.
- At startup, each Datanode connects to its corresponding Namenode and does handshaking.
- Verification of namespace ID and software version of DataNode take place by handshaking.
- At the time of mismatch found, DataNode goes down automatically.

Hadoop Distributed File System (HDFS)-Cont'd

Tasks of HDFS DataNode:

- DataNode performs operations like block replica creation, deletion, and replication according to the instruction of NameNode.
- DataNode manages data storage of the system.

MapReduce

- Is a software framework to **process** vast amount of structured and unstructured data stored in the HDFS.
- MapReduce programs are **parallel** in nature.
- Very useful for performing large-scale data analysis using multiple machines in the cluster.
- Improves the speed and reliability of processing.

Working of MapReduce

➤ MapReduce works by breaking the processing into two phases:

- Map phase and Reduce phase.

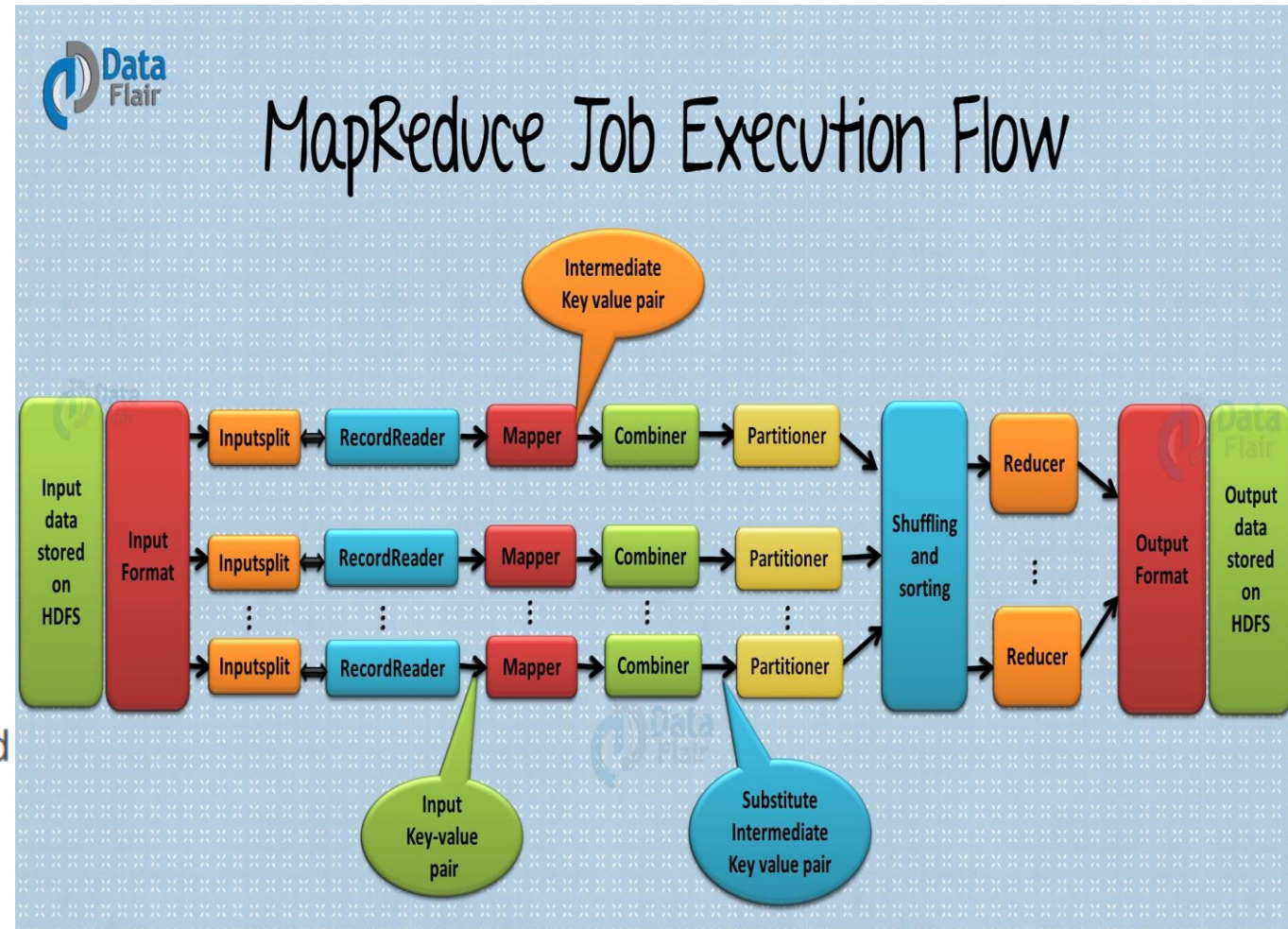
➤ Each phase has key-value pairs as input and output.

◆ The Mapper:

- Each block is processed in isolation by a map task called mapper
- Map task runs on the node where the block is stored

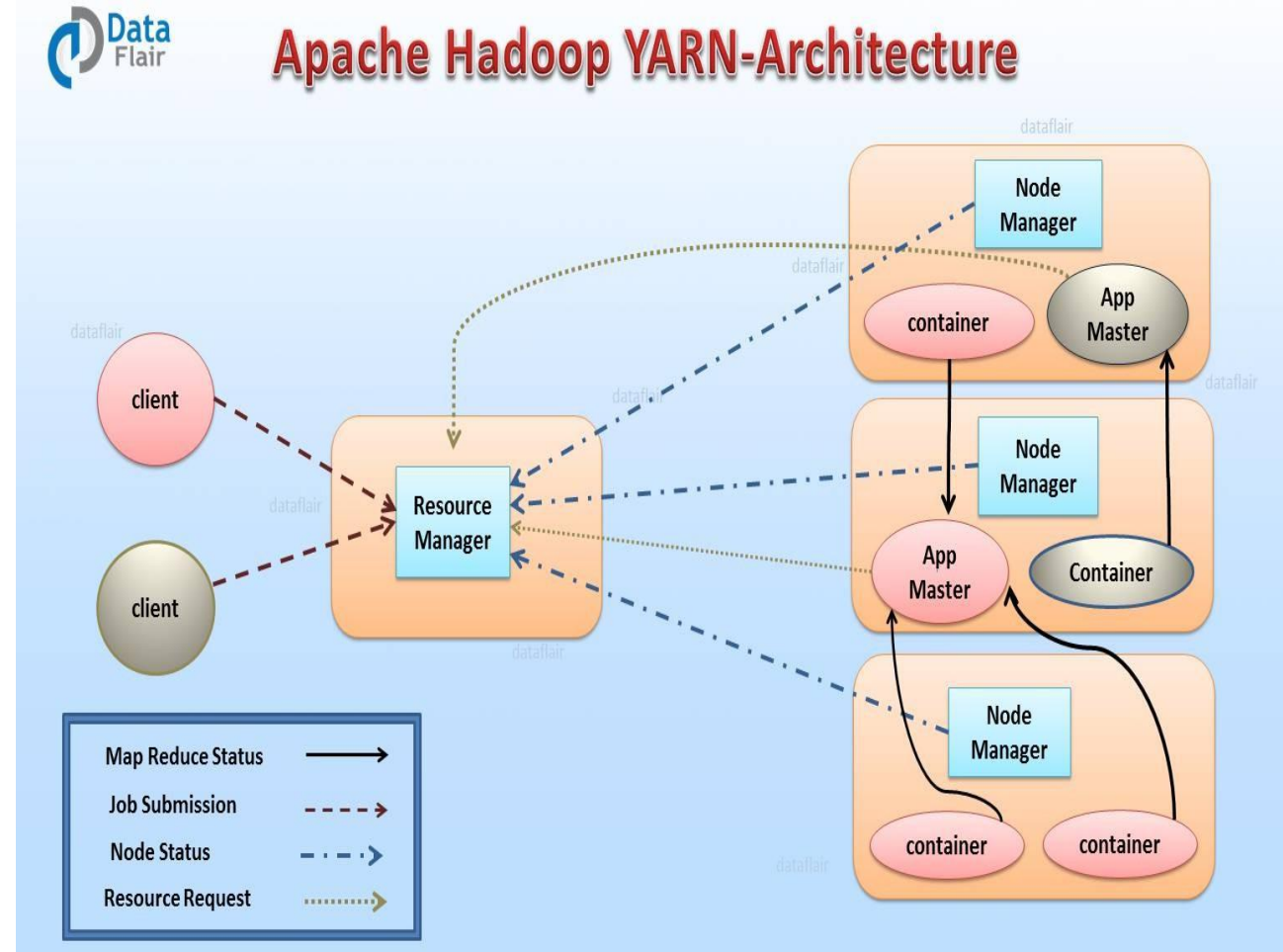
◆ The Reducer:

- Consolidate result from different mappers
- Produce final output



Yet Another Resource Negotiator (YARN)

- Provides the **resource management**.
- YARN is called as the **operating system** of Hadoop as it is responsible for managing and monitoring workloads.
- It allows multiple data processing engines real-time streaming and batch processing to handle data stored on a single platform.



Other Hadoop components in Ecosystem

HBase	Hadoop database for random read/write access
Hive	SQL-like queries and tables on large datasets
Pig	Data flow language and compiler
Oozie	Workflow for interdependent Hadoop jobs
Sqoop	Integration of databases and data warehouses with Hadoop
Flume	Configurable streaming data collection
ZooKeeper	Coordination service for distributed applications

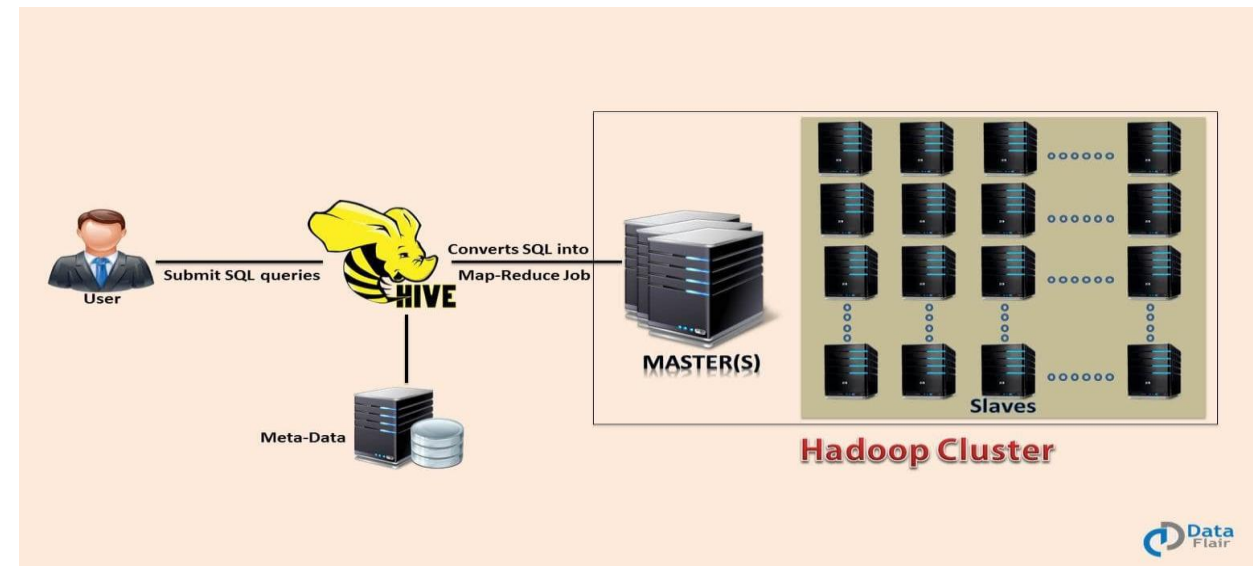
Hive

➤ Data warehouse system for **querying and analyzing** large datasets stored in Hadoop files.

➤ Hive do three main functions:

- ✓ *Data summarization.*
- ✓ *Data query.*
- ✓ *Data analysis.*

➤ Uses language called HiveQL (HQL), similar to SQL.



- ◆ An sql like interface to Hadoop.
- ◆ Data warehouse infrastructure built on top of Hadoop
- ◆ Provide data summarization, query and analysis
- ◆ Query execution via MapReduce
- ◆ Hive interpreter convert the query to Map reduce format.

HBase

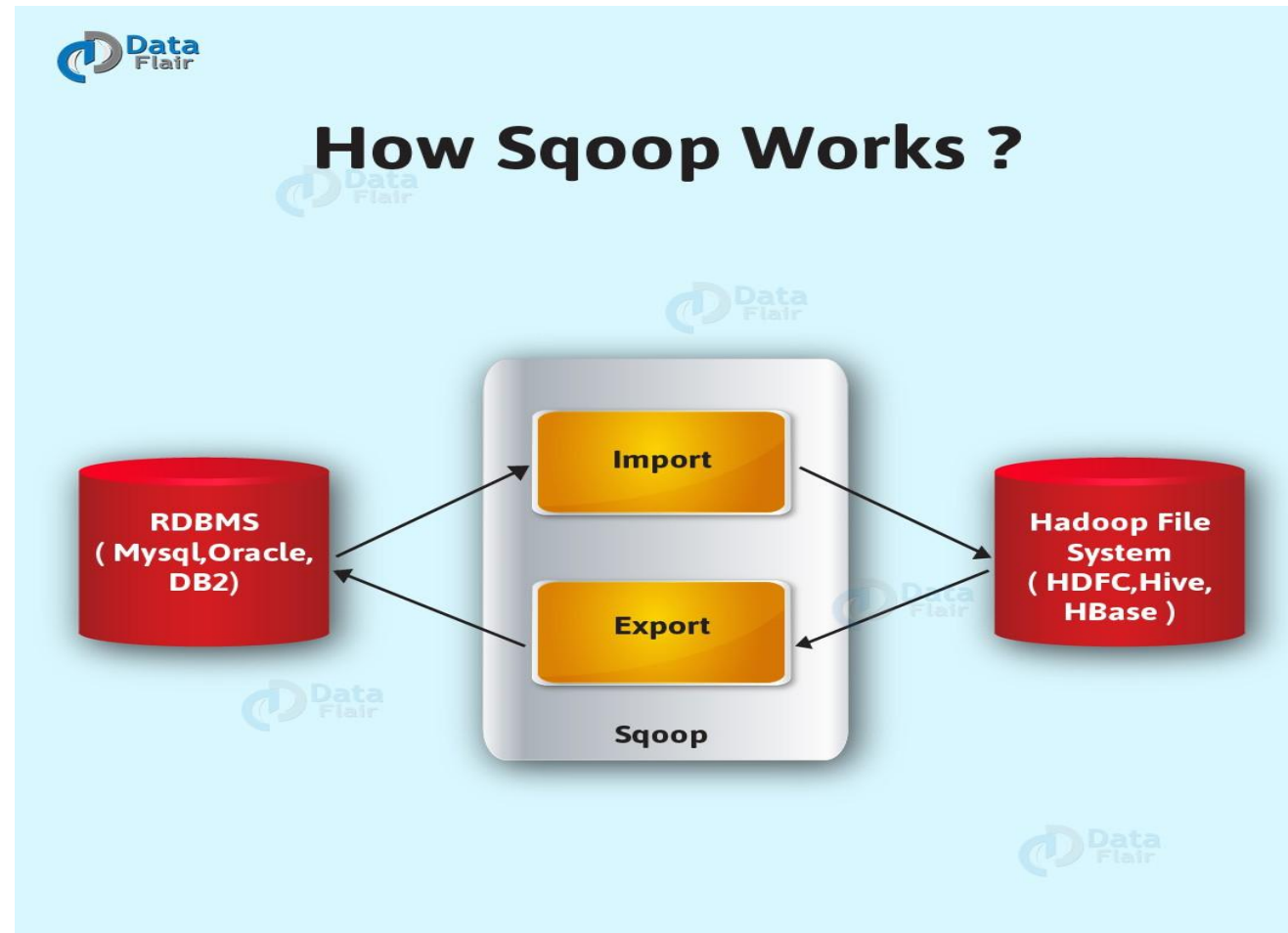
- Is a distributed column-oriented database that was designed to store data in tables that could have billions of row and millions of columns.
- HBase is scalable, distributed, and NoSQL database that is built on top of HDFS.
- HBase, provide real-time access to read or write data in HDFS.

Mahout

- Open-source framework for creating **scalable machine learning** algorithm and data mining library.
- Once data is stored in Hadoop HDFS, mahout provides the data science tools to automatically find meaningful patterns:
 - ✓ **Clustering** – takes the item and organizes them into naturally occurring groups, such that item belonging to the same group are similar to each other.
 - ✓ **Collaborative filtering** – mines user behavior and makes product recommendations (e.g. Amazon recommendations)
 - ✓ **Classifications** – learns from existing categorization and then assigns unclassified items to the best category.
 - ✓ **Frequent pattern mining** – analyzes items in a group (e.g. items in a shopping cart or terms in query session) and then identifies which items typically appear together.

Sqoop

- **Imports** data from external sources into related Hadoop ecosystem components like HDFS, HBase or Hive.
- It also exports data from Hadoop to other external sources.
- Sqoop works with relational databases such as teradata, Netezza, oracle, MySQL.



- ◆ Command-line interface for transforming data between relational database and Hadoop

Flume

- Apache Flume is a distributed, reliable, and available service designed for efficiently **collecting**, **aggregating**, and **transporting** large amounts of streaming data into the Hadoop Distributed File System (HDFS).
- Flume supports various data sources, including
 - log files, social media feeds, databases, and custom applications.
- Flume operates on an event-based model, where data is treated as a stream of events.

What is NoSQL (Not Only SQL)?

- Database management system that designed to handle large volumes of **unstructured** or **semi-structured** data.
- NoSQL databases are often used in big data and real-time web applications due to their flexibility, scalability, and performance.
- Many NoSQL databases are built on distributed architectures, which provide redundancy and fault tolerance, ensuring high availability of data.
- Store and manage unstructured data, such as documents, images, and videos.
- Types of NoSQL Databases:
 - Document Stores: E.g. MongoDB
 - Key-Value Stores: E.g. Redis, Amazon DynamoDB
 - Column-Family Stores: E.g. Apache Cassandra, Hbase
 - Graph Databases: E.g. Neo4j, Amazon Neptune

Hardware Specification for Hadoop

≡ Big Data Management 10.2.1 Performance Tuning and Sizing Guidelines

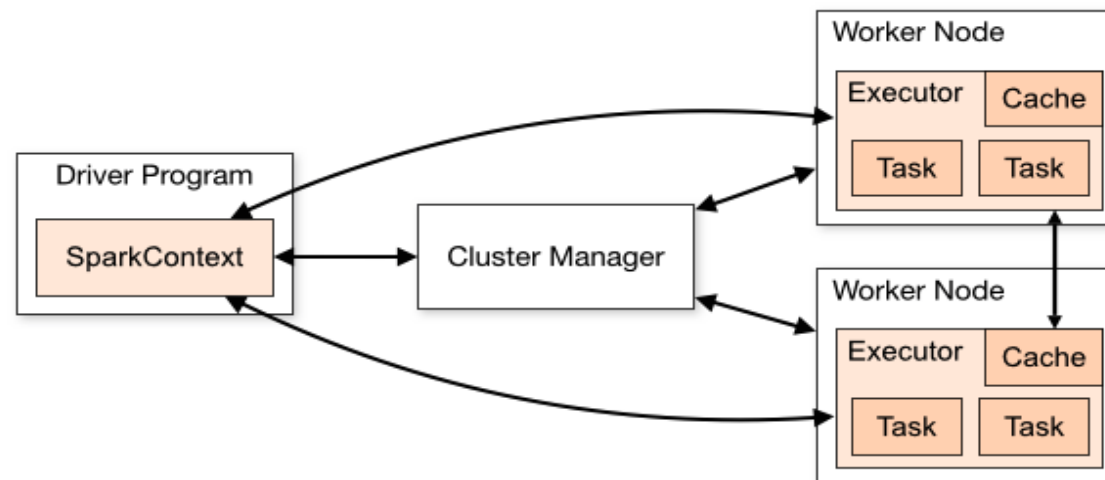
Hardware	Sandbox Deployment	Basic or Standard Deployment	Advanced Deployment
CPU speed	2 - 2.5 GHz	2 - 2.5 GHz	2.5 - 3.5 GHz
Logical or virtual CPU cores	16	24 - 32	48
Total system memory	16 GB	64 GB	128 GB
Local disk space for yarn.nodemanager.local-dirs ¹	256 GB	500 GB	2.4 TB
DFS block size	128 MB	256 MB	256 MB
HDFS replication factor	3	3	3
Disk capacity	32 GB	256 GB - 1 TB	1.2 TB
Total number of disks for HDFS	2	8	12
Total HDFS capacity per node	64 GB	2 - 8 TB	At least 14 TB
Number of nodes	2 +	4 - 10+	12 +
Total HDFS capacity on the cluster	128 GB	8 - 80 TB	144 TB
Actual HDFS capacity (with replication)	43 GB	2.66 TB	57.6 TB
/tmp mount point	20 GB	20 GB	30 GB
Installation disk space requirement	12 GB	12 GB	12 GB



What is Spark

- Apache Spark is an open-source (originally developed at UC Berkeley in 2009), distributed processing system used for big data workloads.
- It utilizes in-memory caching, and optimizes query execution for fast analytic queries for big data.
- Spark supports multiple languages with built in APIs in Java, Scala, Python and R.

The driver/worker design



What is Spark cont'd

- Spark is built on Scala, but has wrappers in python. This wrapper is intuitively called PySpark.
- The two base data structures in Spark:
 - RDD (resilient distributed dataset): completely unstructured and are used for low level api's with transformations and actions.
 - DataSets/DataFrames: is more akin to a pandas dataframe and offers more structure such as having named columns.

Working with PySpark DataFrames

- **SparkConf** sets up the spark session configuration.
- **SparkContext** is the main Spark application.
- The driver program use the SparkContext to **connect** and **communicate** with the cluster
- it helps in executing and coordinating the Spark job with the resource manager(YARN)
- Using SparkContext you can get access to other contexts like **SQL Context** and **Hive Context**.
- Hive Context and SQL Context provided access to these storage types/commands.
- SQL Context is your gateway to Spark SQL and it enables applications to run SQL queries programmatically. Once you have the SQL Context you can start working with **dataframe**, dataset etc.
- Hive Context is your gateway to Hive. It is a super set of the SQLContext i.e. it has support for the functionalities of SQL Context plus hive specific functionalities like access to Hive UDFs, and the ability to **read** data from Hive tables.

Hadoop Vs Spark

Hadoop	Spark
Primarily uses the MapReduce model for batch processing.	Designed for both batch processing and real-time stream processing.
Requires writing complex MapReduce jobs, which can be more challenging for developers.	Offers higher-level APIs in languages like Scala, Java, Python, and R, making it easier to develop applications.
Uses HDFS for storage. Data is stored in a distributed manner across the cluster.	Can use HDFS, but also integrates with other storage systems.
Fault tolerance through data replication in HDFS and re-executing failed tasks in MapReduce.	Uses a concept called Resilient Distributed Datasets (RDDs), which automatically recovers lost data by keeping track of the lineage of transformations.
Ideal for batch processing tasks, data archiving, and ETL (Extract, Transform, Load) processes.	Suited for real-time data processing, machine learning, graph processing, and interactive queries.

Google Colab

- To run Spark we will use Google Colaboratory (Colab), which provides Google-hosted, cloud-based notebooks.
- Google Colab is essentially a Jupyter notebook that runs in the cloud. You can write and execute code, and the results are displayed within the notebook.
- We can access it from [Google Colaboratory](#)
- We need google account to use google colab

How to start Google Colab

- Navigate to Google drive and select Go to Google Drive.
- Click the New button and select “Folder” to create a new folder. Name the folder.
- From the new folder download/connect google colab application following these steps
 - Click New
 - Scroll down to “More” and expand the drop down menu
 - At the bottom of the menu, click “Connect more apps.”
 - Type “colab” in the top-right search field and press enter to search for the Colaboratory application.
 - Click the Connect button to download the Colaboratory application.
- Next, we can create a Colab Notebook by clicking “New” followed by “More,” and then selecting “Colaboratory.”